

1. Experiment Overview

This report compares execution pass-rates across three evaluation batches of the SkillsBench prefill-context pipeline. All batches use Claude Sonnet 4.6 with prefilled skill documents (SKILL.md) selected by embedding retrieval. Each trial runs the agent in a Docker sandbox with Bash/Read/Write/Edit tools and validates via pytest. First20 contains 20 tasks but only 14 are evaluable on host (6 have heavy Dockerfiles); see 20-Task Report v4 for paired analysis.

Batch Summary

Batch	Tasks	Trials	Valid	Rate-Lim	Pass	Pass Rate	Source
First20	20	2333	2330	3	404	17.3%	sb_prefill_n5.jsonl
Smoke10	10	720	719	1	300	41.7%	smoke10_n5.jsonl
Smoke20	20	1500	1500	0	321	21.4%	smoke20_n5.jsonl
ALL	50	4553	4549	4	1025	22.5%	--

2. Per-Task Pass Rates

All 50 tasks sorted by pass rate within each batch. Green $\geq 70\%$, Yellow 30-70%, Red $< 30\%$.

First20 (20 tasks, original eval)

Batch	Task	Pass/Total	Rate	Avg Wall	Pool Sizes	Noise Modes
First20	dialogue-parser	47/48	97.9%	265s	5,10,20,50	easy,hard,random
First20	earthquake-plate-calculation	42/48	87.5%	96s	5,10,20,50	easy,hard,random
First20	citation-check	41/48	85.4%	172s	5,10,20,50	easy,hard,random
First20	econ-detrending-correlation	41/48	85.4%	157s	5,10,20,50	easy,hard,random
First20	earthquake-phase-association	37/48	77.1%	651s	5,10,20,50	easy,hard,random
First20	energy-market-pricing	45/118	38.1%	218s	5,10,20,50	easy,hard,random
First20	adaptive-cruise-control	18/48	37.5%	823s	5,10,20,50	easy,hard,random
First20	exceltable-in-ppt	17/48	35.4%	128s	5,10,20,50	easy,hard,random
First20	dapt-intrusion-detection	47/183	25.7%	29s	5,10,20,50	easy,hard,random
First20	3d-scan-calc	47/588	8.0%	15s	5,10,20,50	easy,hard,random
First20	energy-ac-optimal-power-flow	14/189	7.4%	106s	5,10,20,50	easy,hard,random
First20	enterprise-information-search	3/48	6.2%	306s	5,10,20,50	easy,hard,random
First20	civ6-adjacency-optimizer	5/579	0.9%	61s	5,10,20,50	easy,hard,random
First20	azure-bgp-oscillation-route-leak	0/48	0.0%	542s	5,10,20,50	easy,hard,random
First20	court-form-filling	0/48	0.0%	N/A	5,10,20,50	easy,hard,random
First20	crystallographic-wyckoff-position-analysis	0/48	0.0%	N/A	5,10,20,50	easy,hard,random
First20	data-to-d3	0/48	0.0%	N/A	5,10,20,50	easy,hard,random
First20	dynamic-object-aware-egomotion	0/48	0.0%	N/A	5,10,20,50	easy,hard,random
First20	edit-pdf	0/48	0.0%	N/A	5,10,20,50	easy,hard,random
First20	exoplanet-detection-period	0/4	0.0%	N/A	5	easy,random

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Smoke10 (10 tasks)

Batch	Task	Pass/Total	Rate	Avg Wall	Pool Sizes	Noise Modes
Smoke10	hvac-control	68/75	90.7%	193s	5,10,20,50,100	easy,hard,random
Smoke10	software-dependency-audit	58/75	77.3%	390s	5,10,20,50,100	easy,hard,random
Smoke10	mario-coin-counting	43/75	57.3%	221s	5,10,20,50,100	easy,hard,random
Smoke10	flood-risk-analysis	38/75	50.7%	372s	5,10,20,50,100	easy,hard,random
Smoke10	pptx-reference-formatting	38/75	50.7%	480s	5,10,20,50,100	easy,hard,random
Smoke10	sec-financial-report	32/75	42.7%	248s	5,10,20,50,100	easy,hard,random
Smoke10	r2r-mpc-control	20/75	26.7%	627s	5,10,20,50,100	easy,hard,random
Smoke10	jpg-ocr-stat	2/52	3.8%	885s	5,10,20,50,100	easy,hard,random
Smoke10	financial-modeling-qa	1/75	1.3%	193s	5,10,20,50,100	easy,hard,random
Smoke10	glm-lake-mendota	0/68	0.0%	593s	5,10,20,50,100	easy,hard,random

Smoke20 (20 tasks)

Batch	Task	Pass/Total	Rate	Avg Wall	Pool Sizes	Noise Modes
Smoke20	offer-letter-generator	69/75	92.0%	161s	5,10,20,50,100	easy,hard,random
Smoke20	sales-pivot-analysis	68/75	90.7%	342s	5,10,20,50,100	easy,hard,random
Smoke20	grid-dispatch-operator	60/75	80.0%	221s	5,10,20,50,100	easy,hard,random
Smoke20	lake-warming-attribution	43/75	57.3%	72s	5,10,20,50,100	easy,hard,random
Smoke20	powerlifting-coef-calc	25/75	33.3%	263s	5,10,20,50,100	easy,hard,random
Smoke20	fix-druid-loophole-cve	23/75	30.7%	859s	5,10,20,50,100	easy,hard,random
Smoke20	travel-planning	16/75	21.3%	448s	5,10,20,50,100	easy,hard,random
Smoke20	protein-expression-analysis	14/75	18.7%	404s	5,10,20,50,100	easy,hard,random
Smoke20	reserves-at-risk-calc	3/75	4.0%	833s	5,10,20,50,100	easy,hard,random
Smoke20	video-filler-word-remover	0/75	0.0%	459s	5,10,20,50,100	easy,hard,random
Smoke20	weighted-gdp-calc	0/75	0.0%	309s	5,10,20,50,100	easy,hard,random
Smoke20	setup-fuzzing-py	0/75	0.0%	867s	5,10,20,50,100	easy,hard,random
Smoke20	fix-build-agentops	0/75	0.0%	503s	5,10,20,50,100	easy,hard,random
Smoke20	pg-essay-to-audiobook	0/75	0.0%	718s	5,10,20,50,100	easy,hard,random
Smoke20	xlsx-recover-data	0/75	0.0%	226s	5,10,20,50,100	easy,hard,random
Smoke20	python-scala-translation	0/75	0.0%	895s	5,10,20,50,100	easy,hard,random
Smoke20	shock-analysis-demand	0/75	0.0%	843s	5,10,20,50,100	easy,hard,random
Smoke20	shock-analysis-supply	0/75	0.0%	900s	5,10,20,50,100	easy,hard,random
Smoke20	fix-build-google-auto	0/75	0.0%	594s	5,10,20,50,100	easy,hard,random
Smoke20	simpo-code-reproduction	0/75	0.0%	185s	5,10,20,50,100	easy,hard,random

3. Unified Task Ranking (All 50 Tasks)

All tasks ranked by pass rate, color-coded. Tasks with 0% are listed at bottom.

#	Batch	Task	Pass/Total	Rate	Median Wall
1	First20	dialogue-parser	47/48	97.9%	263s
2	Smoke20	offer-letter-generator	69/75	92.0%	147s
3	Smoke10	hvac-control	68/75	90.7%	177s
4	Smoke20	sales-pivot-analysis	68/75	90.7%	292s
5	First20	earthquake-plate-calculation	42/48	87.5%	89s
6	First20	citation-check	41/48	85.4%	159s
7	First20	econ-detrending-correlation	41/48	85.4%	128s
8	Smoke20	grid-dispatch-operator	60/75	80.0%	158s
9	Smoke10	software-dependency-audit	58/75	77.3%	359s
10	First20	earthquake-phase-association	37/48	77.1%	688s
11	Smoke10	mario-coin-counting	43/75	57.3%	188s
12	Smoke20	lake-warming-attribution	43/75	57.3%	66s
13	Smoke10	flood-risk-analysis	38/75	50.7%	257s
14	Smoke10	pptx-reference-formatting	38/75	50.7%	443s
15	Smoke10	sec-financial-report	32/75	42.7%	183s
16	First20	energy-market-pricing	45/118	38.1%	1s
17	First20	adaptive-cruise-control	18/48	37.5%	900s
18	First20	exceltable-in-ppt	17/48	35.4%	107s
19	Smoke20	powerlifting-coef-calc	25/75	33.3%	235s
20	Smoke20	fix-druid-loophole-cve	23/75	30.7%	900s
21	Smoke10	r2r-mpc-control	20/75	26.7%	775s
22	First20	dapt-intrusion-detection	47/183	25.7%	1s
23	Smoke20	travel-planning	16/75	21.3%	451s
24	Smoke20	protein-expression-analysis	14/75	18.7%	355s
25	First20	3d-scan-calc	47/588	8.0%	1s

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Unified Task Ranking (cont.)

#	Batch	Task	Pass/Total	Rate	Median Wall
26	First20	energy-ac-optimal-power-flow	14/189	7.4%	1s
27	First20	enterprise-information-search	3/48	6.2%	301s
28	Smoke20	reserves-at-risk-calc	3/75	4.0%	900s
29	Smoke10	jpg-ocr-stat	2/52	3.8%	900s
30	Smoke10	financial-modeling-qa	1/75	1.3%	172s
31	First20	civ6-adjacency-optimizer	5/579	0.9%	1s
32	First20	azure-bgp-oscillation-route-leak	0/48	0.0%	489s
33	First20	court-form-filling	0/48	0.0%	N/A
34	First20	crystallographic-wyckoff-position-analysis	0/48	0.0%	N/A
35	First20	data-to-d3	0/48	0.0%	N/A
36	First20	dynamic-object-aware-egomotion	0/48	0.0%	N/A
37	First20	edit-pdf	0/48	0.0%	N/A
38	First20	exoplanet-detection-period	0/4	0.0%	N/A
39	Smoke10	glm-lake-mendota	0/68	0.0%	630s
40	Smoke20	video-filler-word-remover	0/75	0.0%	391s
41	Smoke20	weighted-gdp-calc	0/75	0.0%	298s
42	Smoke20	setup-fuzzing-py	0/75	0.0%	900s
43	Smoke20	fix-build-agentops	0/75	0.0%	431s
44	Smoke20	pg-essay-to-audiobook	0/75	0.0%	838s
45	Smoke20	xlsx-recover-data	0/75	0.0%	208s
46	Smoke20	python-scala-translation	0/75	0.0%	900s
47	Smoke20	shock-analysis-demand	0/75	0.0%	900s
48	Smoke20	shock-analysis-supply	0/75	0.0%	900s
49	Smoke20	fix-build-google-auto	0/75	0.0%	566s
50	Smoke20	simpo-code-reproduction	0/75	0.0%	170s

4. Pass Rate by Pool Size x Noise Mode

Factorial grid showing pass rate per (pool_size, noise_mode) combination. Each cell shows pass/total (rate%). Compares how difficulty scaling affects each batch differently.

First20 Grid

Pool\Noise	easy	hard	random
5	31/318 (10%)	33/195 (17%)	35/192 (18%)
10	28/256 (11%)	32/76 (42%)	32/109 (29%)
20	33/196 (17%)	33/316 (10%)	34/159 (21%)
50	35/76 (46%)	40/226 (18%)	38/214 (18%)
100	--	--	--

Smoke10 Grid

Pool\Noise	easy	hard	random
5	20/50 (40%)	23/50 (46%)	21/50 (42%)
10	14/45 (31%)	20/45 (44%)	23/49 (47%)
20	18/48 (38%)	18/45 (40%)	19/45 (42%)
50	24/50 (48%)	17/50 (34%)	19/50 (38%)
100	22/45 (49%)	21/48 (44%)	21/50 (42%)

Smoke20 Grid

Pool\Noise	easy	hard	random
5	28/100 (28%)	22/100 (22%)	17/100 (17%)
10	24/100 (24%)	22/100 (22%)	20/100 (20%)
20	24/100 (24%)	19/100 (19%)	20/100 (20%)
50	25/100 (25%)	18/100 (18%)	23/100 (23%)
100	20/100 (20%)	18/100 (18%)	21/100 (21%)

5. Difficulty Analysis

Pass Rate vs Pool Size (aggregated across noise modes)

Batch	Pool Size	Pass/Total	Rate
First20	5	99/705	14.0%
First20	10	92/441	20.9%
First20	20	100/671	14.9%
First20	50	113/516	21.9%

Batch	Pool Size	Pass/Total	Rate
Smoke10	5	64/150	42.7%
Smoke10	10	57/139	41.0%
Smoke10	20	55/138	39.9%
Smoke10	50	60/150	40.0%
Smoke10	100	64/143	44.8%

Batch	Pool Size	Pass/Total	Rate
Smoke20	5	67/300	22.3%
Smoke20	10	66/300	22.0%
Smoke20	20	63/300	21.0%
Smoke20	50	66/300	22.0%
Smoke20	100	59/300	19.7%

Pass Rate vs Noise Mode (aggregated across pool sizes)

Batch	Noise Mode	Pass/Total	Rate
First20	easy	127/846	15.0%
First20	hard	138/813	17.0%
First20	random	139/674	20.6%

Batch	Noise Mode	Pass/Total	Rate
Smoke10	easy	98/238	41.2%
Smoke10	hard	99/238	41.6%
Smoke10	random	103/244	42.2%

Batch	Noise Mode	Pass/Total	Rate
Smoke20	easy	121/500	24.2%
Smoke20	hard	99/500	19.8%
Smoke20	random	101/500	20.2%

6. Zero-Pass Tasks Analysis

19 of 50 tasks have 0% pass rate across all conditions.

Batch	Task	Trials	Avg Wall(s)	Pool Sizes	Rate-Lim
First20	azure-bgp-oscillation-route-leak	48	542	5,10,20,50	0
First20	court-form-filling	48	N/A	5,10,20,50	0
First20	crystallographic-wyckoff-position-analysis	48	N/A	5,10,20,50	0
First20	data-to-d3	48	N/A	5,10,20,50	0
First20	dynamic-object-aware-egomotion	48	N/A	5,10,20,50	0
First20	edit-pdf	48	N/A	5,10,20,50	0
First20	exoplanet-detection-period	4	N/A	5	0
Smoke10	glm-lake-mendota	68	593	5,10,20,50,100	0
Smoke20	video-filler-word-remover	75	459	5,10,20,50,100	0
Smoke20	weighted-gdp-calc	75	309	5,10,20,50,100	0
Smoke20	setup-fuzzing-py	75	867	5,10,20,50,100	0
Smoke20	fix-build-agentops	75	503	5,10,20,50,100	0
Smoke20	pg-essay-to-audiobook	75	718	5,10,20,50,100	0
Smoke20	xlsx-recover-data	75	226	5,10,20,50,100	0
Smoke20	python-scala-translation	75	895	5,10,20,50,100	0
Smoke20	shock-analysis-demand	75	843	5,10,20,50,100	0
Smoke20	shock-analysis-supply	75	900	5,10,20,50,100	0
Smoke20	fix-build-google-auto	75	594	5,10,20,50,100	0
Smoke20	simpo-code-reproduction	75	185	5,10,20,50,100	0

Failure Pattern Analysis

Zero-pass tasks are categorized by likely failure mode based on wall time and error patterns. No wall data (N/A) = agent never launched (env/Docker skip). Short wall time (<30s avg) = immediate setup crash. Long wall time (>60s) = execution attempt that does not converge.

No wall data (agent skipped): 6 tasks

Setup/env failures (wall < 30s): 0 tasks

Execution failures (wall >= 30s): 13 tasks

7. Task Difficulty Tiers

Tasks classified into difficulty tiers based on observed pass rate. This helps identify which tasks benefit most from skill prefilling.

Tier	Count	% of 50	Example Tasks
Easy ($\geq 70\%$)	10	20%	dialogue-parser, offer-letter-generator, hvac-control, sales-pivot-analysis, earthquake-plate-calculat...
Medium (30-70%)	10	20%	mario-coin-counting, lake-warming-attribution, flood-risk-analysis, pptx-reference-formatting, sec-financial-report...
Hard (1-30%)	11	22%	r2r-mpc-control, dapt-intrusion-detection, travel-planning, protein-expression-analys, 3d-scan-calc...
Unsolved (0%)	19	38%	azure-bgp-oscillation-rou, court-form-filling, crystallographic-wyckoff-, data-to-d3, dynamic-object-aware-egom...

Easy ($\geq 70\%$)

[First20] dialogue-parser: 47/48 (97.9%)
 [Smoke20] offer-letter-generator: 69/75 (92.0%)
 [Smoke10] hvac-control: 68/75 (90.7%)
 [Smoke20] sales-pivot-analysis: 68/75 (90.7%)
 [First20] earthquake-plate-calculation: 42/48 (87.5%)
 [First20] citation-check: 41/48 (85.4%)
 [First20] econ-detrending-correlation: 41/48 (85.4%)
 [Smoke20] grid-dispatch-operator: 60/75 (80.0%)
 [Smoke10] software-dependency-audit: 58/75 (77.3%)
 [First20] earthquake-phase-association: 37/48 (77.1%)

Medium (30-70%)

[Smoke10] mario-coin-counting: 43/75 (57.3%)
 [Smoke20] lake-warming-attribution: 43/75 (57.3%)
 [Smoke10] flood-risk-analysis: 38/75 (50.7%)
 [Smoke10] pptx-reference-formatting: 38/75 (50.7%)
 [Smoke10] sec-financial-report: 32/75 (42.7%)
 [First20] energy-market-pricing: 45/118 (38.1%)
 [First20] adaptive-cruise-control: 18/48 (37.5%)
 [First20] exceltable-in-ppt: 17/48 (35.4%)
 [Smoke20] powerlifting-coef-calc: 25/75 (33.3%)
 [Smoke20] fix-druid-loophole-cve: 23/75 (30.7%)

Hard (1-30%)

[Smoke10] r2r-mpc-control: 20/75 (26.7%)
 [First20] dapt-intrusion-detection: 47/183 (25.7%)
 [Smoke20] travel-planning: 16/75 (21.3%)
 [Smoke20] protein-expression-analysis: 14/75 (18.7%)
 [First20] 3d-scan-calc: 47/588 (8.0%)
 [First20] energy-ac-optimal-power-flow: 14/189 (7.4%)
 [First20] enterprise-information-search: 3/48 (6.2%)
 [Smoke20] reserves-at-risk-calc: 3/75 (4.0%)
 [Smoke10] jpg-ocr-stat: 2/52 (3.8%)
 [Smoke10] financial-modeling-qa: 1/75 (1.3%)
 [First20] civ6-adjacency-optimizer: 5/579 (0.9%)

Unsolved (0%)

[First20] azure-bgp-oscillation-route-leak: 0/48 (0.0%)
 [First20] court-form-filling: 0/48 (0.0%)

[First20] crystallographic-wyckoff-position-analysis: 0/48 (0.0%)

[First20] data-to-d3: 0/48 (0.0%)

[First20] dynamic-object-aware-egomotion: 0/48 (0.0%)

[First20] edit-pdf: 0/48 (0.0%)

[First20] exoplanet-detection-period: 0/4 (0.0%)

[Smoke10] glm-lake-mendota: 0/68 (0.0%)

[Smoke20] video-filler-word-remover: 0/75 (0.0%)

[Smoke20] weighted-gdp-calc: 0/75 (0.0%)

[Smoke20] setup-fuzzing-py: 0/75 (0.0%)

[Smoke20] fix-build-agentops: 0/75 (0.0%)

[Smoke20] pg-essay-to-audiobook: 0/75 (0.0%)

[Smoke20] xlsx-recover-data: 0/75 (0.0%)

[Smoke20] python-scala-translation: 0/75 (0.0%)

[Smoke20] shock-analysis-demand: 0/75 (0.0%)

[Smoke20] shock-analysis-supply: 0/75 (0.0%)

[Smoke20] fix-build-google-auto: 0/75 (0.0%)

[Smoke20] simpo-code-reproduction: 0/75 (0.0%)

8. Data Integrity & Reliability Verification

File Checksums (SHA-256, first 16 hex)

sb_prefill_n5.jsonl	55d929f4ca87cafd
smoke10_n5.jsonl	3df086ecb92b2389
smoke20_n5.jsonl	46b6a8f888d4dcf2

Trial Count Verification

First20: 2333 trials across 20 tasks (avg 116.7 trials/task)
 Smoke10: 720 trials across 10 tasks (avg 72.0 trials/task)
 Smoke20: 1500 trials across 20 tasks (avg 75.0 trials/task)

Per-Task Trial Count Range

First20: min=4, max=588, median=48
 Smoke10: min=52, max=75, median=75
 Smoke20: min=75, max=75, median=75

Note: First20 has uneven trial counts (4-588/task) due to incremental Phase P0 seed replication. Smoke20 has 75 trials/task but includes 30 design skips at pool_size=5 for 2 tasks whose |GT|=6 > pool_size (python-scala-translation, travel-planning). Smoke10 is incomplete for 2 tasks: jpg-ocr-stat (52/75) and glm-lake-mendota (68/75) due to resume with an 8-task subset after the initial 10-task run.

Trial ID Uniqueness

Unique config IDs: 3136
 Configs run multiple times: 42
 Extra rows from re-runs: 1417

Note: trial_id encodes (task, pool_size, noise, seed) -- it is a config ID, not a unique execution ID. Repeated IDs are valid independent replications of the same configuration (all in First20's Phase P0). Each row is a distinct agent execution. No cross-batch duplicates exist.

Task Disjointness

First20 & Smoke10: empty (OK)
 First20 & Smoke20: empty (OK)
 Smoke10 & Smoke20: empty (OK)

Rate-Limit Contamination

Trials flagged as rate-limited (stderr mentions rate_limit/overloaded/529, or agent exited immediately with rc=2):

First20: 3/2333 (0.1%)
 Smoke10: 1/720 (0.1%)
 Smoke20: 0/1500 (0.0%)

Verdict: rate-limit contamination is negligible (<0.2%).

Seed Coverage

First20: seeds = [1, 2, 3, 4]
 Smoke10: seeds = [0, 1, 2, 3, 4]
 Smoke20: seeds = [0, 1, 2, 3, 4]

Note: First20 uses seeds [1,2,3,4] (seed 0 was in the earlier Phase C run, stored separately in sb_exec_20t.jsonl). Smoke batches use [0,1,2,3,4].

Model Consistency

claude-sonnet-4-6: 4279 trials
 (no model field): 274 trials from two distinct causes:
 - First20: 244 trials across 6 tasks with heavy Dockerfiles (unsafe for host execution). Agent never launched.

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- Smoke20: 30 trials at pool_size=5 for python-scala-translation and travel-planning, where |GT|=6 > pool_size. These are design skips (GT could not fit in pool). Note: travel-planning is 16/75 overall (21.3%), NOT 0% -- only its pool_size=5 slice was skipped.

Wall-Time Statistics (excluding N/A)

First20: mean=118s, median=1s, p5=1s, p95=786s (n=2089, 244 trials missing wall data)

Smoke10: mean=404s, median=276s, p5=105s, p95=900s (n=720, 0 trials missing wall data)

Smoke20: mean=502s, median=404s, p5=76s, p95=900s (n=1470, 30 trials missing wall data)

Reliability Verdict

PASS with caveats. All three batches use disjoint task sets with no cross-contamination. Rate-limit interference is negligible (<0.2%). Model is consistent (claude-sonnet-4-6) across all executed trials.

DESIGN CAVEATS:

1. First20 has uneven replication (4-588 trials/task) from incremental Phase P0 runs. The 6 tasks with heavy Dockerfiles (court-form-filling, crystallographic-wyckoff, data-to-d3, dynamic-object-aware-egomotion, edit-pdf, exoplanet-detection-period) were never executed on host. Per the 20-Task Report v4, only 14 of these 20 tasks are evaluable.
2. Smoke10 is incomplete for 2 tasks: jpg-ocr-stat (52/75) and glm-lake-mendota (68/75) due to resume with an 8-task subset.
3. Smoke20 row count is balanced (75/task) but includes 30 design skips at pool_size=5 for python-scala-translation and travel-planning where |GT|=6 > pool_size. travel-planning has 16/75 pass overall (21.3%).
4. Per the 20-Task Report v4 paired analysis: aggregate pass-rate differences between prefill conditions are NOT statistically significant ($p > 0.05$ for all comparisons, $N=14$ tasks). Signal is driven by a few standout tasks, not a consistent cross-task effect. These aggregate comparisons cannot support causal claims.